

Introduction to Electronic Design Automation

Homework 3

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1 BDD Operation

1. (a)

$$\begin{aligned}
 & \text{robdd_build}(ab(\neg c + d) + (a\neg b + \neg ab)(c\neg d + \neg cd), 1) \\
 & \xrightarrow{\eta} \text{robdd_build}(b(\neg c + d) + \neg b(c\neg d + \neg cd), 2) \\
 & \xrightarrow{\eta} \text{robdd_build}(\neg c + d, 3) \\
 & \xrightarrow{\eta} \text{robdd_build}(d, 4) \\
 & \xrightarrow{\eta} \text{robdd_build}(1, 5) \\
 & \quad v_1 \\
 & \xrightarrow{\lambda} \text{robdd_build}(0, 5) \\
 & \quad v_0 \\
 & \quad v_2 = (d, v_1, v_0) \\
 & \xrightarrow{\lambda} \text{robdd_build}(1, 4) \\
 & \quad v_1 \\
 & \quad v_3 = (c, v_2, v_1) \\
 & \xrightarrow{\lambda} \text{robdd_build}(c\neg d + \neg cd, 3) \\
 & \xrightarrow{\eta} \text{robdd_build}(\neg d, 4) \\
 & \xrightarrow{\eta} \text{robdd_build}(0, 5) \\
 & \quad v_0 \\
 & \xrightarrow{\lambda} \text{robdd_build}(1, 5) \\
 & \quad v_1 \\
 & \quad v_5 = (d, v_0, v_1) \\
 & \xrightarrow{\lambda} \text{robdd_build}(d, 4) \\
 & \quad v_2 = (d, v_1, v_0) \\
 & \quad v_4 = (c, v_5, v_2) \\
 & \quad v_6 = (c, v_3, v_4) \\
 & \xrightarrow{\lambda} \text{robdd_build}(b(c\neg d + \neg cd), 2) \\
 & \xrightarrow{\eta} \text{robdd_build}(c\neg d + \neg cd, 3) \\
 & \quad v_4 = (c, v_5, v_2) \\
 & \xrightarrow{\lambda} \text{robdd_build}(0, 3) \\
 & \quad v_0 \\
 & \quad v_7 = (b, v_4, v_0) \\
 & \quad v_8 = (a, v_6, v_7)
 \end{aligned}$$

1. (b)

2 SAT Solving

3 Combinational Equivalence Checking

4 Characteristic Function

5 Sequential Equivalence Checking